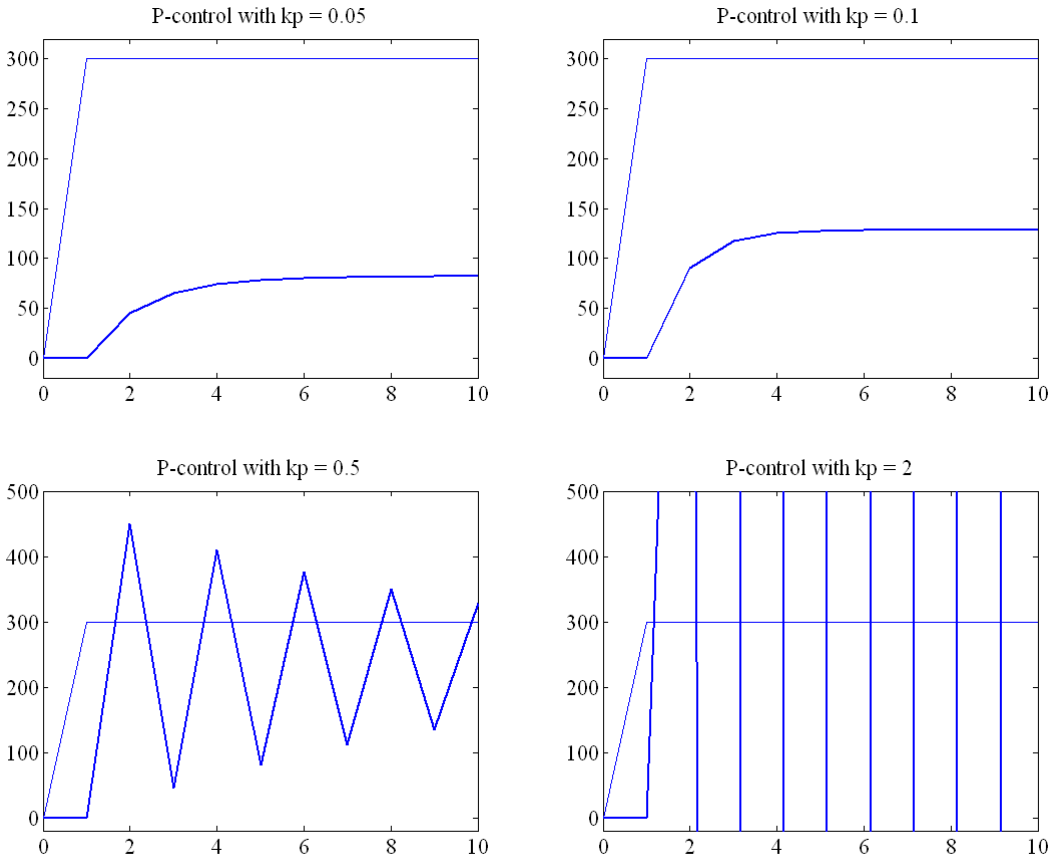
Laboratory 3 – Solutions
Discrete-Time Control System Design using SIMULINK

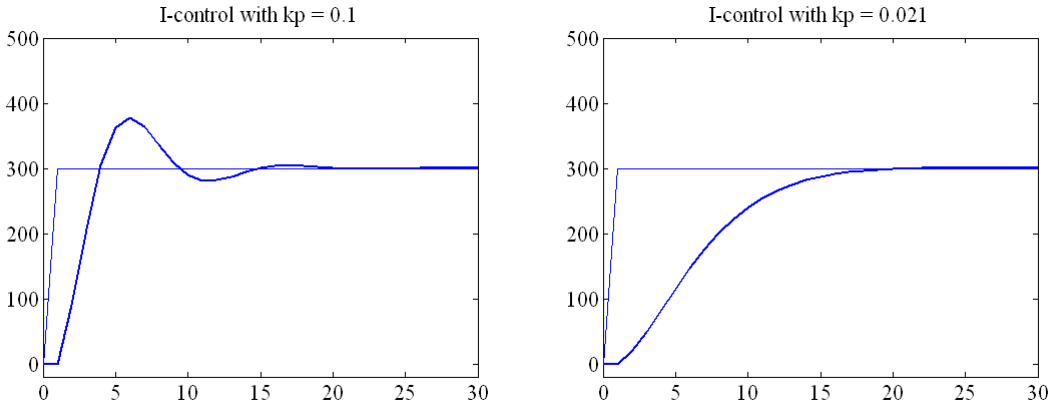
Brief solutions are given below so that answers can be checked. Please contact the lecturer if more information is required.

Exercise 3.1 Proportional control of ventilation rate



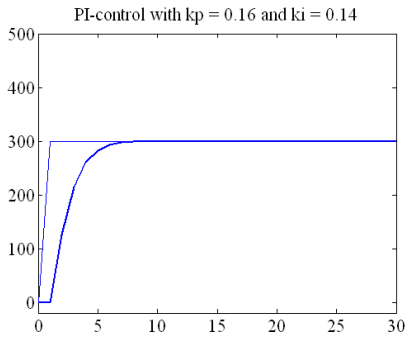
Larger control gains yield a faster response and lower steady state error (if the closed-loop system remains stable). However, if the gain is too high then the closed-loop becomes unstable. Finally, larger control gains yield higher amplitudes for the control input signal (not plotted above) which may also be undesirable in practice – *you should always examine the control input by adding a suitably connected Scope to the Simulink diagram*.

Exercise 3.2 Integral control of ventilation rate

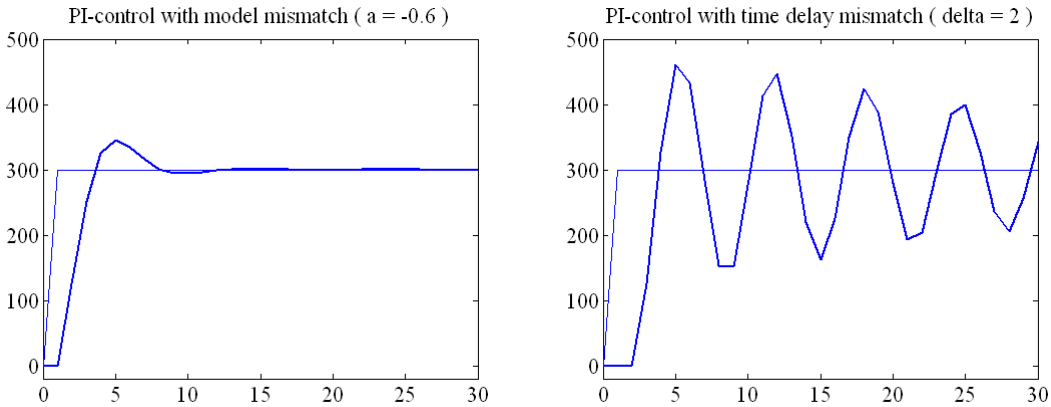


Larger control gains yield a faster response, at the risk of overshoot and oscillations. If the control gain is too large the closed-loop system may become unstable. Note that the left-hand side plot above shows a *stable* response since the oscillations decay over time. Stability and damping (poorly damped systems oscillate) are different matters, do not mix them up! With integral control, steady state tracking is always achieved if the closed-loop system is stable. For this example, in order to avoid the problematic oscillations, we require a slower response speed than that of the proportional controller – note the increased simulation length of 30 seconds, compared to 10 seconds in the earlier graphs.

Exercise 3.3 Proportional-Integral control of ventilation rate

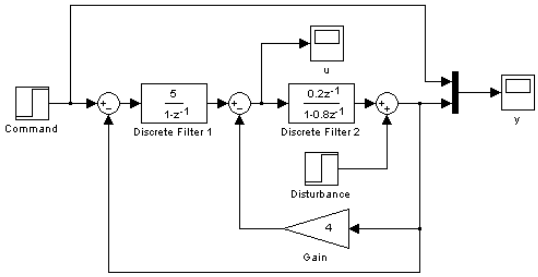


Exercise 3.4 Control of ventilation rate with model mismatch

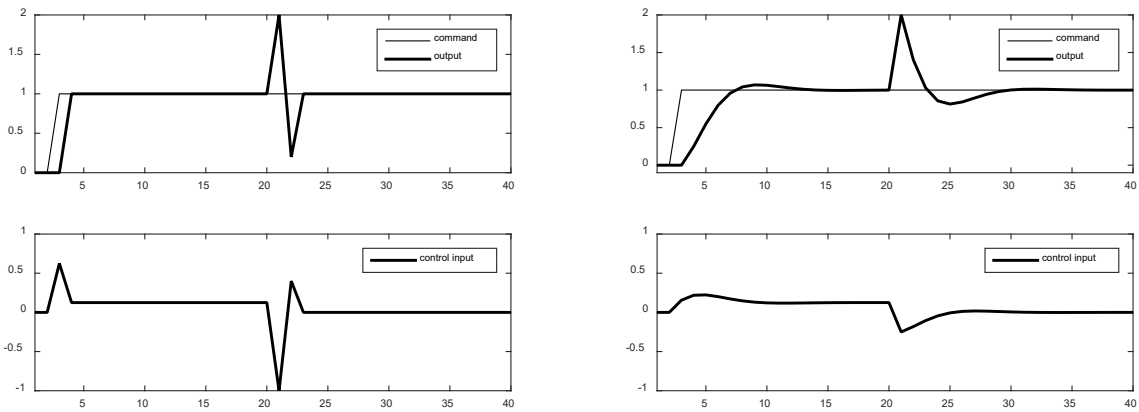


Exercise 3.5 Pole Assignment Design

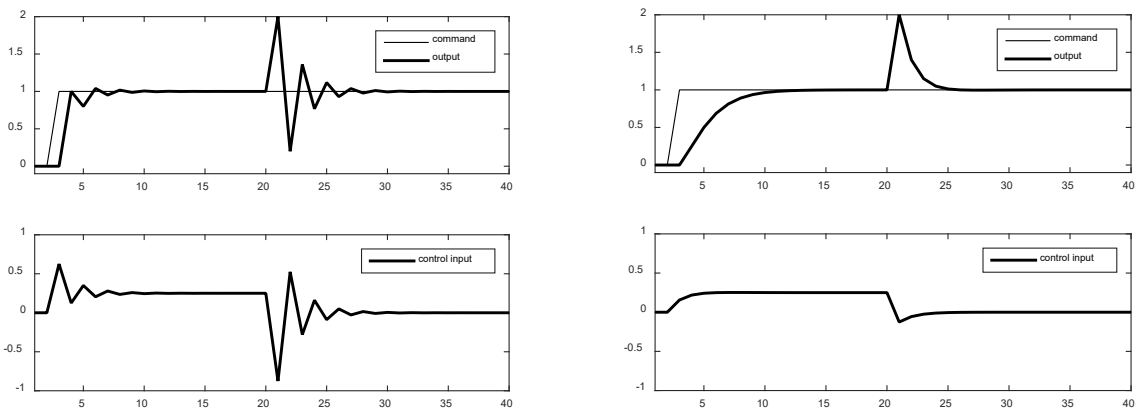
Below is a Simulink diagram showing where to connect the simulated disturbances. This is an example of an output (or load) disturbance. Another type of disturbance would be at the input. The numerical values in this figure have been copied from another document and relate to the reservoir model and not the axial fan! Hence, your solution needs to use the values quoted in the laboratory notes.



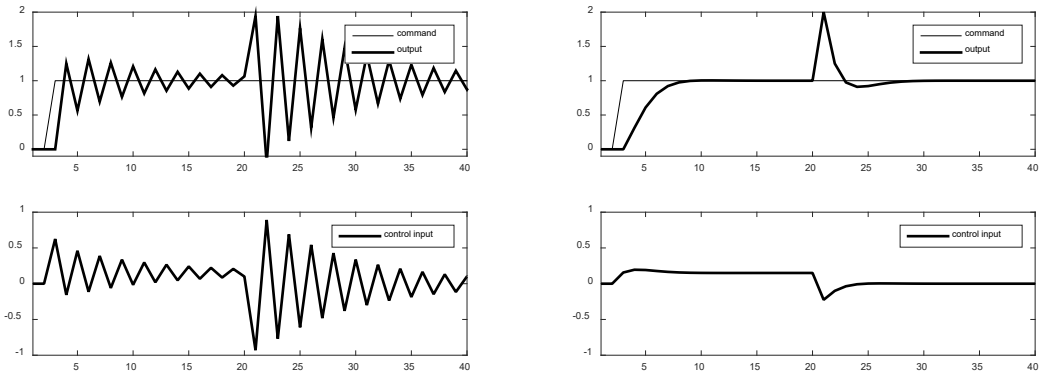
In the figures below, the deadbeat response with output disturbances is shown on the left-hand side and the alternative design on the right hand side.



In the figures below, the deadbeat response with model mismatch ($a_1 = -0.7$) is shown on the left-hand side and the alternative design on the right hand side.



In the figures below, the deadbeat response with model mismatch ($a_1 = -0.7$ and $b_1 = 2.0$) is shown on the left hand side and the alternative design again on the right hand side.



The right-hand side plots above show how powerful a well-designed feedback controller can be. Good control is obtained despite severe disturbances and significant modelling errors! The disadvantage of deadbeat control is a qualitative answer that can be interpreted from these and related plots.